

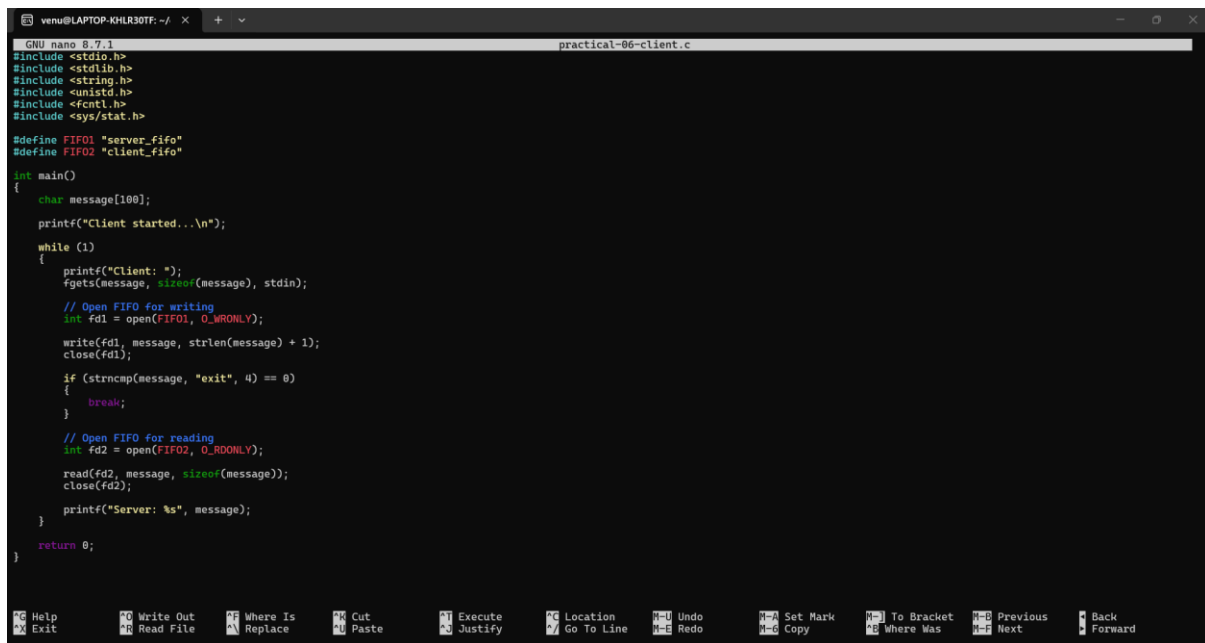
PRACTICAL-06

Create a client-server application using Named Pipes (FIFOs). The client sends messages and the server processes and responds to them. Analyze FIFO behavior when multiple clients communicate simultaneously.

Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSR1. Demonstrate asynchronous event handling and explain the role of signal handlers.

CODE:-

Client code:-



```
GNU nano 2.7.1 practical-06-client.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>

#define FIFO1 "server_fifo"
#define FIFO2 "client_fifo"

int main()
{
    char message[100];

    printf("Client started...\n");

    while (1)
    {
        printf("Client: ");
        fgets(message, sizeof(message), stdin);

        // Open FIFO for writing
        int fd1 = open(FIFO1, O_WRONLY);

        write(fd1, message, strlen(message) + 1);
        close(fd1);

        if (strcmp(message, "exit", 4) == 0)
        {
            break;
        }

        // Open FIFO for reading
        int fd2 = open(FIFO2, O_RDONLY);

        read(fd2, message, sizeof(message));
        close(fd2);

        printf("Server: %s", message);
    }

    return 0;
}
```

Server code:-

```
GNU nano 8.7.1 practical-06-server.c
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>

#define FIFO1 "server_fifo"
#define FIFO2 "client_fifo"

int main()
{
    char message[100];

    // Create named pipes
    mkfifo(FIFO1, 0660);
    mkfifo(FIFO2, 0660);

    printf("Server started...\n");
    printf("Waiting for client...\n");

    while (1)
    {
        // Open FIFO for reading
        int fd1 = open(FIFO1, O_RDONLY);

        read(fd1, message, sizeof(message));
        close(fd1);

        printf("Client: %s\n", message);

        // Exit condition
        if (strcmp(message, "exit") == 0)
        {
            printf("Client disconnected.\n");
            break;
        }

        // Prepare response
        printf("Server: *\n");
        fgets(message, sizeof(message), stdin);

        // Open FIFO for writing
        int fd2 = open(FIFO2, O_WRONLY);

        write(fd2, message, strlen(message) + 1);
        close(fd2);

        unlink(FIFO1);
        unlink(FIFO2);
    }

    return 0;
}
```

Signal code:-

```
GNU nano 8.7.1 practical-06-signal.c
#include <stdio.h>
#include <stdlib.h>
#include <signal.h>
#include <unistd.h>

void signal_handler(int sig)
{
    if (sig == SIGINT)
    {
        printf("\nSIGINT received.\n");
        printf("Interrupt signal received from keyboard.\n");
    }
    else if (sig == SIGTERM)
    {
        printf("\nSIGTERM received.\n");
        printf("Termination signal received.\n");
    }
    else if (sig == SIGUSR1)
    {
        printf("\nSIGUSR1 received.\n");
        printf("User-defined signal received.\n");
    }
}

int main()
{
    [ Read 41 lines ]
    ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
    ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line  M-E Redo      M-6 Copy
```

OUTPUT:-

```

cd ~/os-labcd ~/os-lab siddhartha@SIDDHARTHAVARMA:~$ cd ~/os-lab
siddhartha@SIDDHARTHAVARMA:~/os-lab$ cd practical
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ pwd
/home/venu/os-lab/practical
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ cd practical-06
-bash: cd: practical-06: Not a directory
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ nano practical-06-server.c
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ nano practical-06-client.c
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ gcc practical-06-server.c -o server
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ gcc practical-06-client.c -o client
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ ls
client          practical       practical-03.c practical-04.c practical-05.c practical-06-client.c practical-06.c program-01.c program-02.c server
destination.txt practical-03    practical-04    practical-05    practical-06    practical-06-server.c program        program-02    program.c    source.txt
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ ./server
Server started...
Waiting for client...
Client: Hello Server

Server: ^C
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ nano practical-06-signal.c
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ gcc practical-06-signal.c -o signal
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ ls
client          practical       practical-03.c practical-04.c practical-05.c practical-06-client.c practical-06.c program        program-02.c server_fifo
client_fifo     practical-03    practical-04.c practical-06    practical-06-server.c practical-06-signal.c program-01.c program.c    signal
destination.txt practical-03.c  practical-05    practical-06-client.c practical-06.c    program-02    server        source.txt
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ ./signal
Process ID: 841
Waiting for signals...
^C
SIGINT received.
Interrupt signal received from keyboard.

```

```

siddhartha@SIDDHARTHAVARMA:~$ kill -USR1 1234
-bash: kill: (1234) - No such process
siddhartha@SIDDHARTHAVARMA:~$ cd os-lab
siddhartha@SIDDHARTHAVARMA:~/os-lab$ cd practical
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ cd singal
-bash: cd: singal: No such file or directory
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ ^C
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ ^C
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ practical-06-signal.c
practical-06-signal.c: command not found
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ ./signal
Process ID: 956
Waiting for signals...
|

```

```
venu@LAPTOP-KHLR30TF: ~/ × + ▼  
siddhartha@SIDDHARTHAVARMA:~$ cd ~/os-lab/practical  
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ ./client  
Client started...  
Client: Hello Server  
exit  
^C  
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ [ ]
```

```
venu@LAPTOP-KHLR30TF: ~/ × + ▼  
siddhartha@SIDDHARTHAVARMA:~$ cd ~/os-lab/practical  
siddhartha@SIDDHARTHAVARMA:~/os-lab/practical$ ./client  
Client started...  
Client: hello  
|
```